

ELEVATE 1.0



Team Name: Team Preflight

Product: Preflight. Should you believe this ML paper?

PS ID : EL-01

PS Name : AI-Powered ML Paper Reproducibility Platform

Abstract :

Preflight re-runs every number in an ML paper over ten seeds, explains each gap with a counterfactual re-run, and flags results inflated by leakage, unfair baselines, wrong metrics or lucky seeds.

Claude reads the paper; re-runs decide. 13/13 planted issues caught, 0 false alarms.



[Open the live demo](#)

PROPOSED SOLUTION



The problem: ML results can reproduce and still mislead: leaked test data, untuned baselines, lucky seeds.

Today's tools stop at “does it run?”. Preflight asks “should you believe it?”



What EL-01 asks for	What Preflight does
Understand the methodology	AI reader extracts claims and stated settings; you confirm
Find the implementation and config	README, config and AST mapping down to file:line
Execute in a controlled environment	Sandbox: browser WASM, subprocess or Docker; 10 seeds
Compare with reported results	Seed-aware verdict with a k-seed tolerance
Explain why results differ	Shapley gap attribution and missing-detail inference
Go beyond code exists	Leakage, metric, fair-baseline and fragility re-runs
Evidence-backed, per experiment	Claim Ledger: table cell, code line, command, log, hash
Lightweight CPU-only papers	CPU triage: full, scaled or untestable, with the reason

THE POINT

Reproducing is not enough. We show whether to believe a result, and prove it with a re-run.

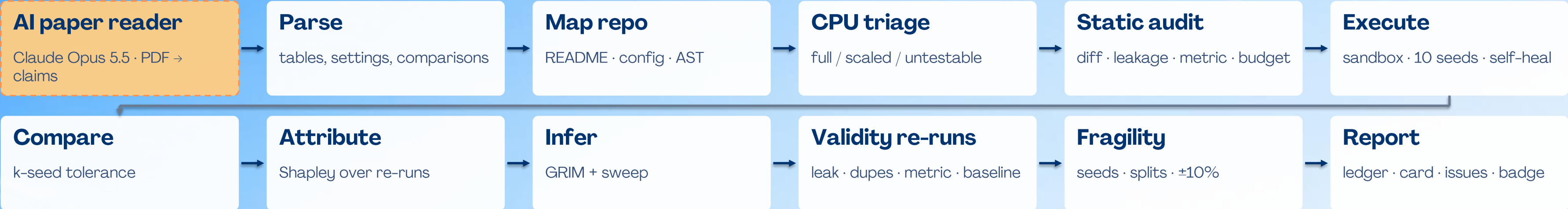
ID	CLAIM	REPORTED → MEASURED	PER SEED	VERDICT
C1	Logistic Regression on Breast Cancer Table 1 · Test accuracy (%) on Breast Cancer. hyperparameter mismatch	98.25 → 95.79 ± 1.23, 10 seeds		Not reproduced
C2	Decision Tree on Breast Cancer Table 1 · Test accuracy (%) on Breast Cancer.	93.86 → 93.60 ± 2.71, 10 seeds		Reproduced
C3	SVM (RBF) on Breast Cancer Table 1 · Test accuracy (%) on Breast Cancer.	98.25 → 97.46 ± 1.13, 10 seeds		Reproduced
C4	k-NN on Breast Cancer Table 1 · Test accuracy (%) on Breast Cancer. seed selection	95.61 → 97.72 ± 0.94, 10 seeds		Not reproduced
C5	ANOVA top-20 + Logistic Regression on SynthGene-2k Table 2 · Test accuracy (%) on SynthGene-2k. data leakage	92.40 → 92.40 ± 6.38, 10 seeds		Reproduced
C6	ViT-B/16 on ImageNet-1k Table 3 · Top-1 accuracy (%) on ImageNet-1k. compute or data limit	81.30 → - not run		Untestable
C7	SVM (RBF) beats k-NN Results text seed selection Fragile Overclaim	2.64 → -0.26 ± 1.38, 10 seeds		Not reproduced

Claim Ledger for Paper A: every number, its verdict and its cause

[Open this ledger live →](#)

TECHNICAL APPROACH

THE POINT
One AI step. Everything after it is deterministic, sandboxed and repeatable.



Dashed = the only AI step (claim extraction, checked by a person). Everything after it is deterministic: static analysis, sandboxed execution, statistics and counterfactual re-runs.

Verdict rule

`tol = max(2σ/√k, 0.5 pt)`
`k = seeds paper averaged`
`Partial: ≤ 3σ/√k`
`or a scaled run`

Gap attribution

$$\phi_i = \sum |S|!(n-|S|-1)!/n! \cdot [v(SU\{i\}) - v(S)]$$

`v(S): 10-seed mean,`
`S set to paper values`

Sandbox

fresh repo copy, hashed
seed set at the code's seed site
Docker: no network, CPU/RAM caps
probe: fits, splits, row hashes

Stack

Python 3.14 · scikit-learn 1.8
Pyodide (WASM), Web Worker
FastAPI · queue · Docker
Claude Opus 5.5, JSON output



The same 11 stages, live in the app

INNOVATION AND UNIQUENESS



	Runs code	Per-claim verdict	Explains why (re-run)	Leakage check	Fragility / seeds	False-alarm control	Metric & baseline audit
VERITAS (2026)	Yes	Yes	Partly	No	No	No	No
CORE-Bench agents	Yes	Partly	No	No	No	No	No
REPRO-Bench agents	Yes	Partly	Partly	No	No	No	No
Code Ocean	Yes	No	No	No	No	No	No
SciCoQA / Dude	No	Partly	Partly	No	No	LLM	No
Preflight	Yes	Yes	Yes	Yes	Yes	Re-run	Yes

Based on each tool's published description. Partly: covers part of the column.

“How is this different from VERITAS?”

VERITAS tells you whether a claim replicates. Preflight tells you why it failed, whether a passing result is inflated, and how fragile it is, and backs each finding with a counterfactual re-run.

Original: a GRIM check rules out split ratios that cannot produce the reported accuracy before anything runs; seeds are injected at the code's own seed site; every flag must survive a re-run (Dude's false-positive problem).

THE POINT

Others say whether. Preflight shows why, and catches results that reproduce but mislead.

96%

of a 2.46-pt gap explained by one config value (C), via Shapley re-runs

-28.8 pt

a result that reproduces, but only because feature selection saw the test set

+1.52 → -0.56

a claimed win that flips once the baseline gets the same tuning budget

15 / 30

conditions where “significantly outperforms” still holds

13 / 13

planted issues caught across the test papers, with 0 false alarms on the clean control. One real 1993 paper audited end to end.

FEASIBILITY AND VIABILITY



THE POINT

Built and running today. Every demo audit finishes within a minute, on a CPU, in the browser.



Every demo audit runs on a CPU in seconds [Run Paper A live](#) →

Paper	Sandboxed runs	From cache	CPU-s (laptop)	Live in browser
A • planted (7 claims)	242	94	3.7	6 s
B • clean control	22	2	1.5	2.6 s
C • planted (6 claims)	151	66	24.4	55 s
R • real, Street et al. 1993	82	49	8.6	13.3 s

Browser: Pyodide 314.0.7 in Chrome. Live and recorded runs match exactly.

Risks and mitigations

Risk	Mitigation
PDF tables misread	Claude with a JSON schema, then a person confirms the claims
LLM hallucination	LLM only extracts; every finding is a re-run
Repo breaks on new libraries	self-heal rules, patch shown, logged as impediment
Experiment too big for CPU	triage: scale rows to the RAM budget, verdict ≤ Partial
Hostile code	Docker: no network, CPU/RAM/PID caps, read-only root
Venue wifi fails	recorded replay, labelled as recorded

Three ways to deploy, one engine

- Browser

Pyodide in a Web Worker. Zero server cost, nothing leaves the laptop. Works today.
- Server

FastAPI, job queue, one Docker container per run, result cache by content hash. Tested.
- CI / author mode

The CLI writes a GitHub Action that re-audits on every push, plus a README badge.

Sustainability

Open-source core (MIT) so labs can self-host and trust the checks. Hosted tier for conferences and journals: batch audits of artifact submissions, badge issuing, reviewer reports.

Compute cost scales with papers audited, and small papers cost nothing because they run in the visitor's browser.

IMPACT AND SCALING

DEP

Who it is for

Reviewers & repro chairs

triage artifact submissions in minutes, with evidence to cite

Students & MLRC

reproduce a paper for coursework and learn why it differs

ML teams

check a paper's claims before building on them

Authors

pre-submission check and a README badge

Tested, not promised

- ✓ 13/13 planted issues caught, 0 false alarms (clean control: 0 findings)
- ✓ 9 automated tests pass: planted, clean, edits, real paper, server
- ✓ Browser and laptop runs give identical numbers
- ✓ Live edits re-audited: fixing C turns C1 green; a new scaler leak is caught
- ✓ Server mode: HTTP submit, poll, idempotent cache, rate limits
- ✓ Real 1993 paper: one claim exact, one flagged, with its reason

THE POINT

Success is cheap. Audits run in the user's browser, so 10× users is mostly static hosting.



ARR

YC says yes tomorrow. Can it handle success next week?

Users

Most audits run in the visitor's browser, so 10× traffic is mostly static file hosting.

Load

Stateless API + queue; autoscaled Docker workers; one container per run.

Repeat work

Runs cached by hash of files, command and library versions; duplicate submissions are free.

Breaks

Per-run CPU, RAM, PID and time caps; network off; rate limit per client.

Crashes

Self-heal known API breaks; failed jobs keep their events; static audit still reports.

Robustness

Idempotent jobs and retries; recorded replay if live compute is down.

RESEARCH AND REFERENCES

THE POINT

Still unsolved. Top agents replicate under a third of papers; top LLMs find under half the discrepancies.



DEP

ARR

21-27%

best agents on PaperBench vs 41% for ML PhDs: replication is still hard (OpenAI, 2025)

46.7%

of real paper–code discrepancies found by the best LLMs (SciCoQA, ACL 2026)

294+

papers in 17 fields affected by leakage (Kapoor & Narayanan, Patterns 2023)

68%

full replication by VERITAS after automated fixes; it reports whether, not why (arXiv 2607.02931)

Try it

Live demo

preflight-el01.github.io/preflight

Code (MIT)

github.com/preflight-el01/preflight

Video

[Watch the 90-second walkthrough](#)

Test kit

[A test paper with planted bugs](#)



In the app: press **Tour** for a guided walkthrough, Ctrl K to search, Audit your own to upload a PDF.

Sources

Kapoor & Narayanan. Leakage and the reproducibility crisis in ML-based science. Patterns, 2023

Starace et al. PaperBench. OpenAI, 2025 · Siegel et al. CORE-Bench, 2024 · REPRO-Bench, ACL 2025

SciCoQA: paper–code alignment, ACL 2026 (arXiv 2601.12910) · Dude, arXiv 2609.03416

VERITAS: a general-purpose replication tool, arXiv 2607.02931

Brown & Heathers. The GRIM test. Social Psychological and Personality Science, 2017

Pineau et al. ML Reproducibility Checklist, 2021 · Kapoor et al. REFORMS, 2023

Street, Wolberg & Mangasarian. Nuclear feature extraction for breast tumor diagnosis. SPIE, 1993

Datasets

Bundled with scikit-learn: Wisconsin Diagnostic Breast Cancer (569 cases), Digits. Generated: SynthGene-2k, SynthMed, SynthHD, SynthKNN-Aug, SynthLarge-2M.

Papers

Three fictional test papers with an answer key (A, B, C) and one real paper (Street et al., 1993). Every number shown comes from a logged sandbox run.

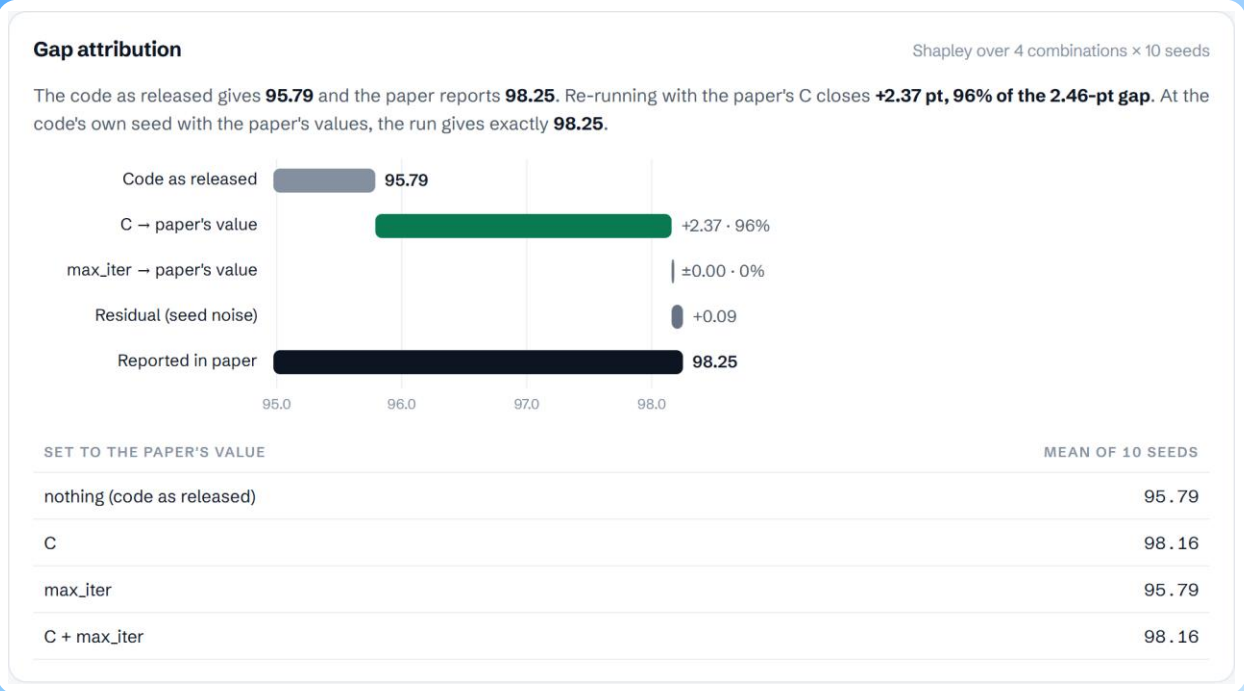
LIVE PROOF



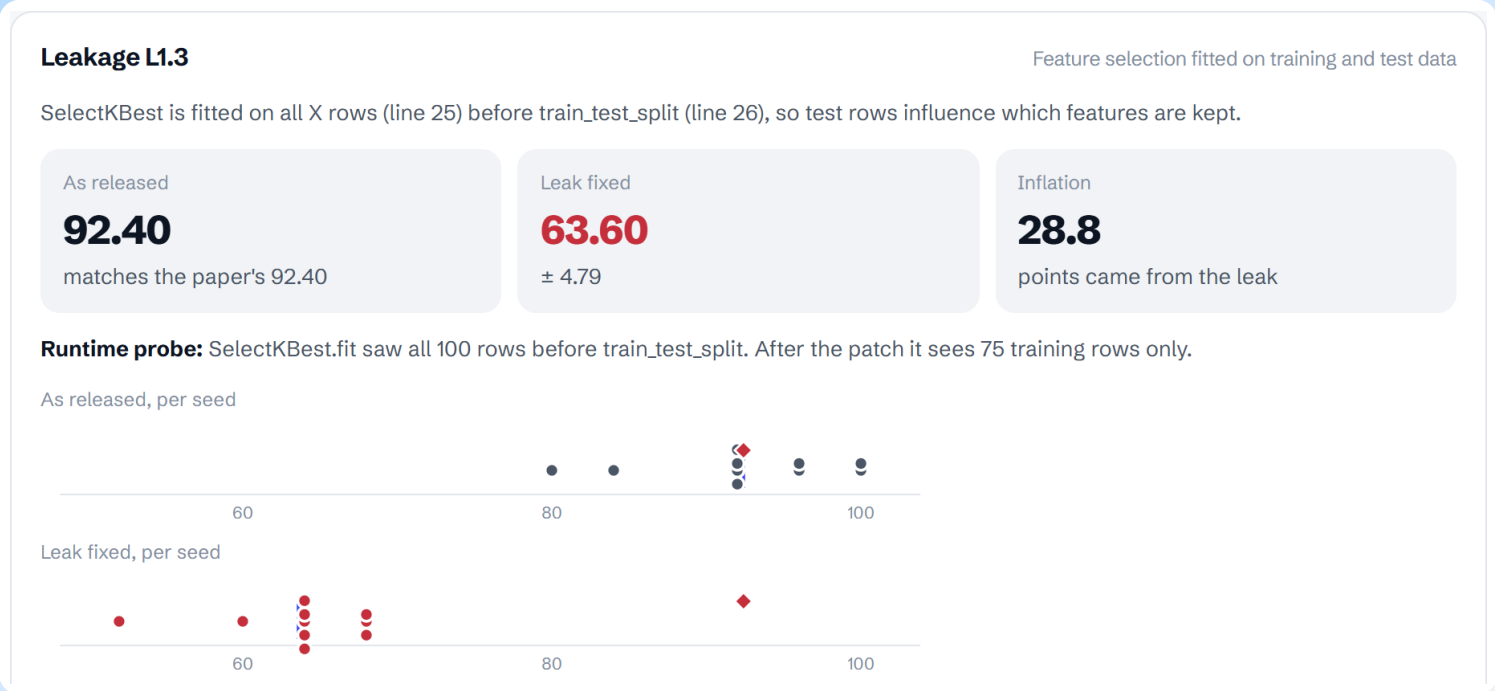
THE POINT

Proof, not promises. Four findings from the real app, each backed by a re-run.

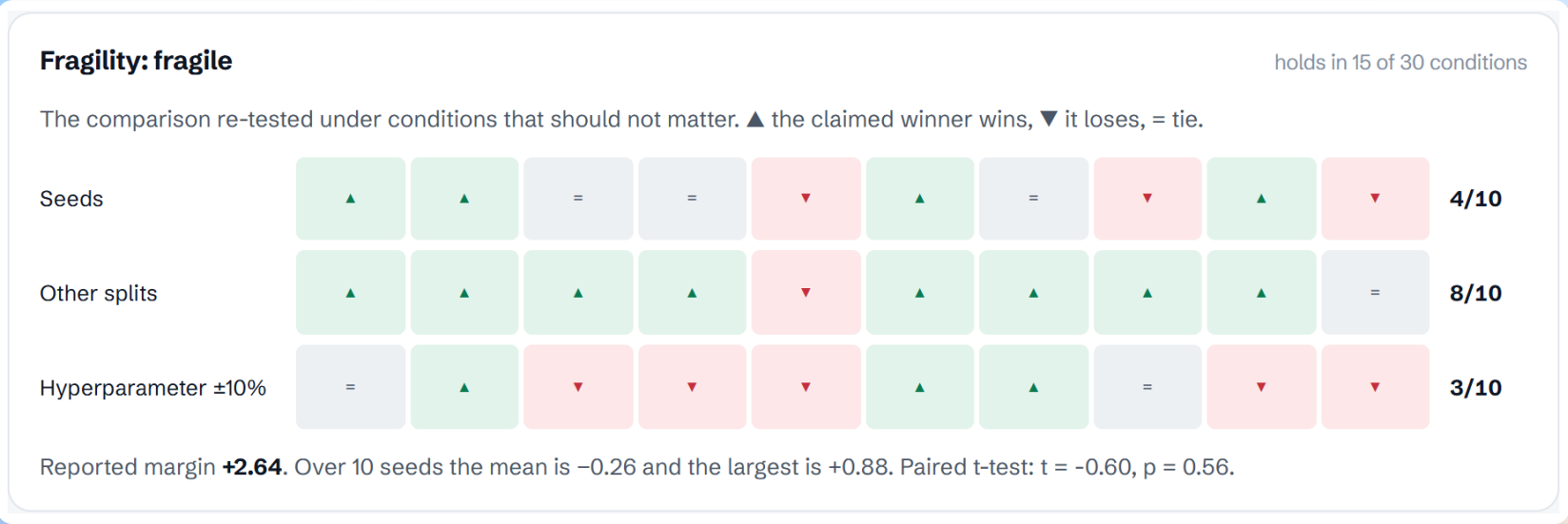
Each screen is a re-run, not an opinion. Click any one to open it live in the app.



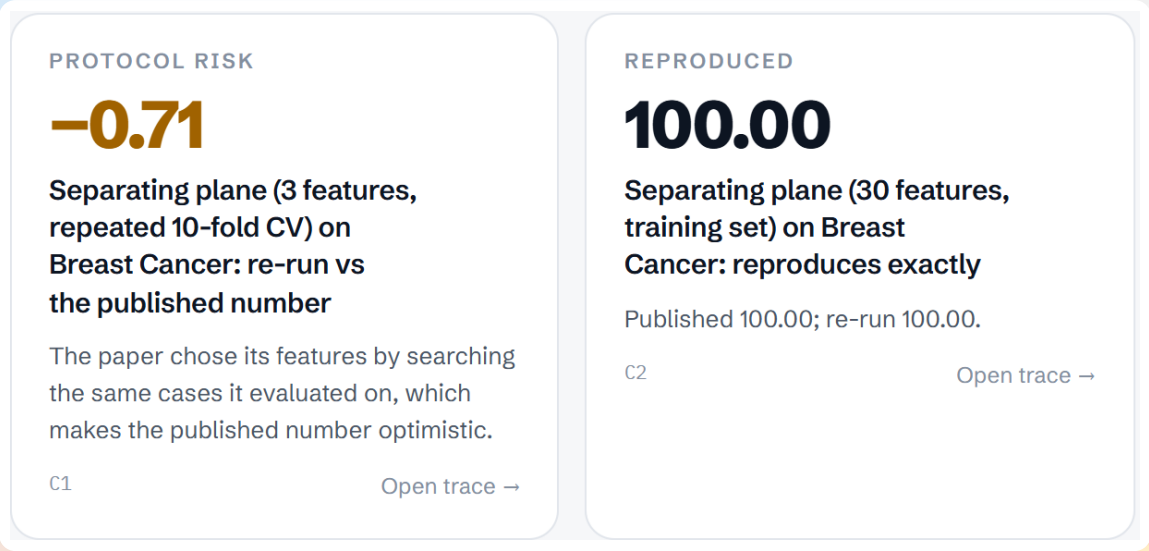
Why C1 fails: one config value explains 96% of the gap →



Why C5 cannot be trusted: the leak, the patch, the drop →



Why C7 is fragile: 15 of 30 conditions →



A real paper: one claim exact, one optimistic, and its reason →

Real paper

Street, Wolberg & Mangasarian, 1993.

The authors released no code, so the repo is our reproduction.